

Internet of Things (IoT)



IoT Ecosystem

Different components of IoT

"Strength is Life, Weakness is Death." – **Swami Vivekananda**

IoT Ecosystem



It encompasses **all the components** needed to collect, process, and analyze data from IoT devices, enabling smart applications and services.

internet connectivity, service platform
including **security**

✓ IoT Core

- Sensors & Actuators, microcontrollers,

✓IoT Gateway

- It ensure bidirectional communication between IoT networks and other networks

✓User Interface / Visualization

✓Cloud

- Design sleek, visually appealing, interactive, and ease-of-use graphical user interface (GUI)

✓IoT Architectures

- **Graphical structure** of the designed IoT-based solutions and products
 - 3-layer architecture, IoT-WF, oneM2M, etc.

- Accepts, accumulates, maintains, stores, and process data

✓Analytics

- It indulges in conversion and **analysis of data** which results in recommendations and future decision making

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IoT Ecosystem



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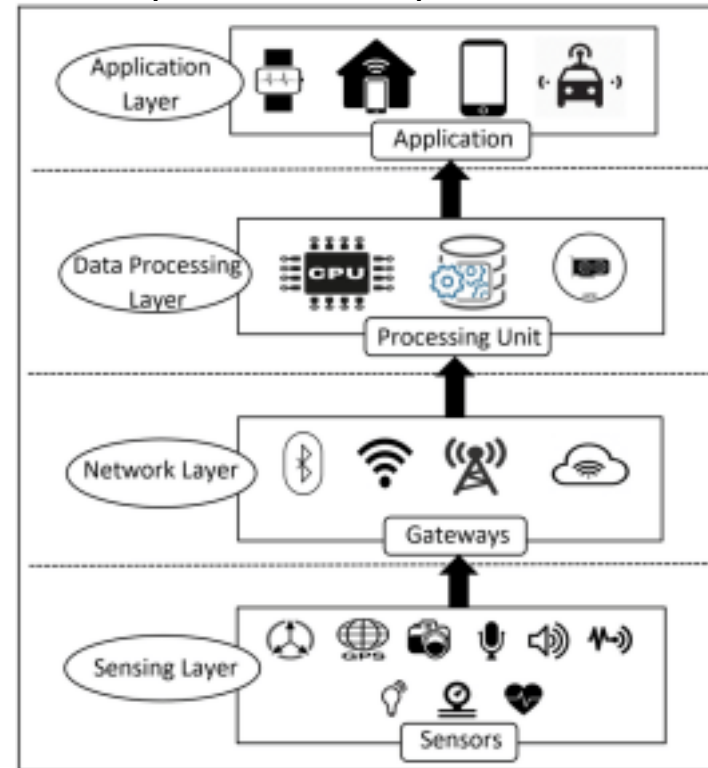
✓ IoT Frameworks

- Tools needed to design and implement IoT-based solutions and products
 - Microsoft Azure IoT,
 - Cisco Ultra IoT,
 - Amazon AWS IoT,
 - IBM Watson IoT,
 - etc.

➤ Network and its application should never be built without careful planning ➤ Architecture is how you design (i.e. [graphical structure](#)) your solution.

➤ IoT architecture refers to the framework that defines [how](#) different components of an IoT system [interact](#) to enable data collection, [transmission](#), [processing](#), and ultimately the [delivery of services](#).

Very common 4-Layer Architecture for IoT



Source: Sikder *et al.*, "A Survey on Sensor-based Threats to Internet-of-Things (IoT) Devices and Applications" *IEEE Communications Surveys & Tutorials*, 23(2), pp. 1125 - 1159, 2021.

Core Components of IoT



Sensors & Actuators

- **Sensors** - to gather data and events

- **Actuators** – responsible for moving and controlling a mechanism or system

information and control command

Platform



- **Service Platform** – ability to **deploy and manage the IoT devices and applications** including data management, data analytics and **security**

Micro

controllers



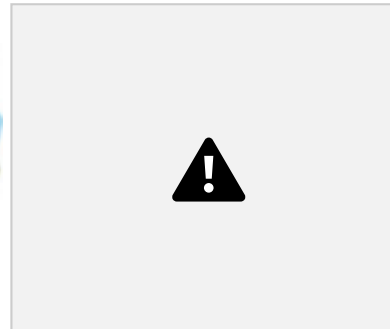
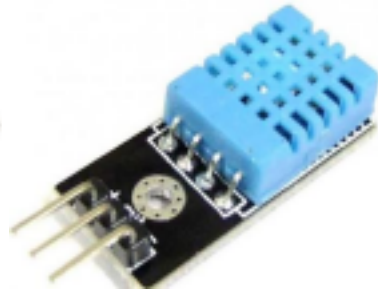
- **Microcontrollers** - automatically controls sensors and actuators; makes them smart

Internet
Connectivity

- **Internet connectivity** – responsible for **sharing**

Service

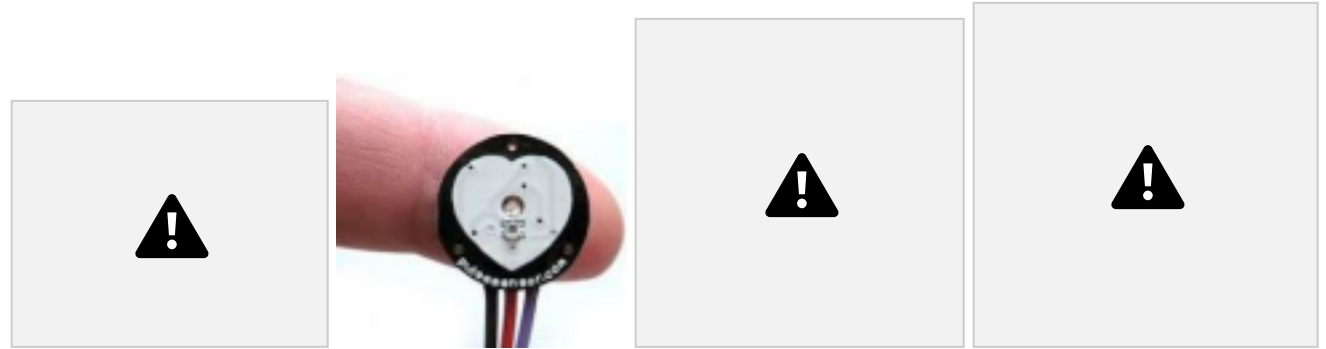
“Things” in IoT – Sensors



MQ135 - Air Quality Gas
Sensor
Sound Detection

Sensor DHT11 - Temperature

and Humidity Sensor
PIR Motion Detector Sensor



Pulse Sensor

Distance Sensor

IR Sensor

LDR Light Sensor Ultrasonic

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“Things” in IoT – Actuators





4 Channel 5V Relay Servo Motor DC Motor Solenoid valve



Linear Actuators LED LCD Diplay

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Access Technologies in IoT

Communication

Criteria ➤ Range

➤ Frequency Bands



- Power Consumption
 - Topology
 - Constrained Devices
 - Constrained-Nodes
- Networks

IoT Access Technologies



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Use of Internet Infrastructure



Key Advantages of IP ➤ Open

IEEE 802.14.4g,

and standard-based ➤ Versatile

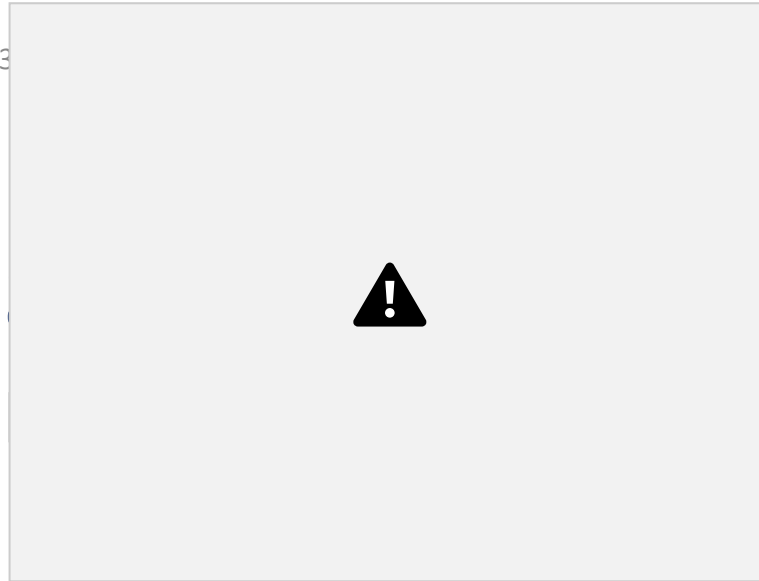
- Ubiquitous
- Scalable
- Manageable
- Highly secure
- Stable and resilient



- IPv6 packets require a minimum MTU/PDU size of 1280 bytes.
- The maximum size of a MAC layer frame in IEEE 802.15.4 is 127 bytes. – It gives just 102 bytes for an IPv6 packet !!

Need of packet/frame size optimization due to

- Constrained Nodes
- Constrained Networks



- IEEE 802.15.4: Low-Rate WPAN
- 6LoWPAN: IPv6 over Low-Power WPAN
- TSCH: Time Synchronized Channel Hopping
- 6TiSCH: IPv6 over the TSCH mode of IEEE 802.15.4e
- 6top: 6TiSCH Operation Sublayer



In 6TiSCH IoT Network



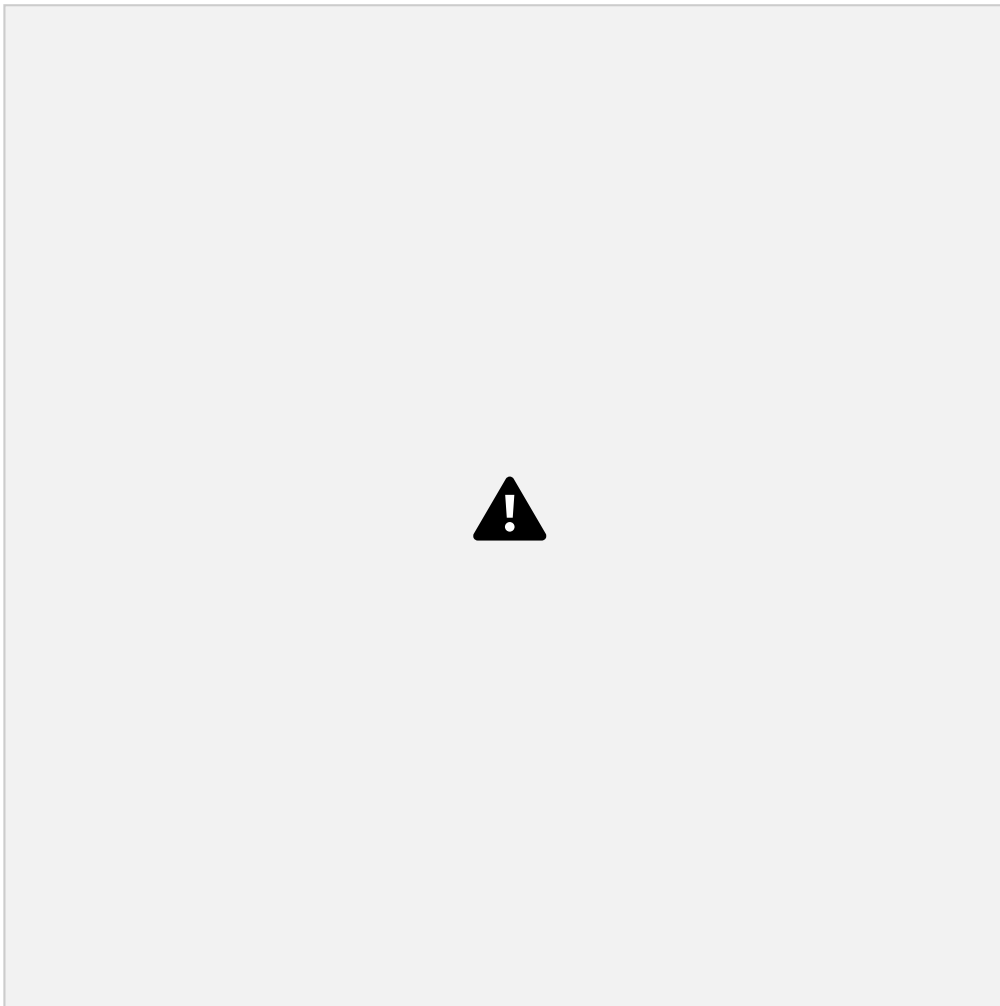
IEEE 802.15.4e -2011
Amendment⁶

- WPAN: Wireless Personal Area Networks

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Application Layer





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IoT Gateway



- It is a **physical device or software program** that serves as the connection point **between the two different types of networks**



Gateway

- Provides bidirectional **communication**
 - Between IoT protocols and other networks
 - e.g. Zigbee <--> Ethernet
- Sometimes programmed to execute some **processing** operations
 - Edge computing
- It is necessary to maintain **security** to a certain extent
 - Can shield the entire IoT systems from any cyberattack

Source: B. Kang, D. Kim, H. Choo, "Internet of Everything: A Large-Scale Autonomic IoT Gateway", IEEE Transactions on Multi-scale Computing Systems, vol. 3, no. 3, 2017, pp. 206-214.

Use of Cloud



- IoT generates vast amount of Big Data; this in turn puts a huge strain on Internet Infrastructure.
- Cloud can facilitate to
 - Provide different services
 - Store huge amount of data
 - Process the data efficiently
- **Benefits of Cloud Platform in IoT**
 - Network Scalability
 - Data Mobility
 - Time to market
 - Security
 - Cost-effectiveness

AI for IoT



- AI focuses on putting human intelligence in machine

- It gives the ability to a machine/program to think and learn by itself

Use of AI in IoT:

- Smart Home

- Automated HVAC control

- Industrial IoT

- Predictive maintenance
 - Optimized supply chain

- Farming

- Smart farming
 - Interruption warning

- Self-driving Car

- Mimic human driving on road



- Health

- Auto-diagnosing any disease –
Assistive healthcare

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Data Analytics in IoT



- ❖ The **business value of IoT** is not just in the ability to **connect devices**, but it comes from **understanding the data** these devices create.

“Data Analytics + IoT => Smart Business Solutions”

- **IoT analytics** is the **application of data analysis tools and procedures** to realize value from the huge volumes of data generated by connected IoT devices

Challenges:

- ✓ Huge Volume

- ✓ Real-time data flow
- ✓ Variety of data types
 - e.g. XML, video, SMS
- ✓ Unstructured data
- ✓ Variable data model and meaning / value

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Securing IoT



- Both the IoT manufacturers and their customers didn't care about the security !

Unauthorized access to IoT devices Unauthorized access to IoT network



Source: <https://www.theguardian.com/technology/2016/oct/26/ddos-attack-dyn-mirai-botnet>

[finland-amidst-winter](#)

Major cyber attack disrupts internet service across Europe and US;
October 26, 2016

DDoS attack halts heating in Finland amidst winter;
November 7, 2016

Source: <http://metropolitan.fi/entry/ddos-attack-halts-heating-in>

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User Interface



• Information made available to the end-users

- Users can actively **check and act in** for their IOT system

Important Characteristics:

- ✓ Sleek design
- ✓ Visually appealing
- ✓ Interactive UI
- ✓ Ease-of-use
- ✓ Handy

Source: <https://www.daikin.com/about/design/2017/05/entry-15>

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IoT Framework



- Framework provides a **development**

environment.

– It provides appropriate infrastructure **to design and implement the architecture**

- IoT framework comprises of **large number of components**

- sensors, sensor systems, gateways, mobile app, embedded controller, data management platform, analytical platform, and so on.

- support **interoperability** among all devices, provides **secure connectivity**, **reliability** in data transfer, **interface** to 3rd party application to built on it, and so on.

RTI (Real-Time Innovations) Connex DDS	Cisco Ultra IoT
Salesforce IoT cloud	Microsoft Azure IoT
Eclipse IoT	PTC ThingWorx
GE (General Electronic) Predix	Amazon AWS IoT
IBM Watson IoT	Kaa

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Lessons Learned



What is IoT Ecosystem

➤ Different components of IoT

- IoT Architecture
- IoT Core
 - ✓ Sensors & Actuators
 - ✓ IoT Access Technologies
 - ✓ Use of Internet
- IoT Gateway
- Use of Cloud in IoT
- AI for IoT
- Data Analytics in IoT
- Security in IoT
- User Interface for IoT
- IoT Framework



